

Total marks – 84

Attempt Questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1	(12 marks)	Marks
(a)	Solve $\frac{4}{x+1} \geq 1$	3
(b)	Evaluate $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$	2
(c)	Let A be the point $(-2, 1)$ and B be the point $(5, 2)$. Find the coordinates of the point P which divides AB externally in the ratio 5:4	2
(d)	Indicate on a number plane the region satisfied by both $x^2 + y^2 \leq 1$ and $y \geq x$	2
(e)	Use the substitution $u^2 = x - 2$ to evaluate $\int_2^3 x\sqrt{x-2} \, dx$	3

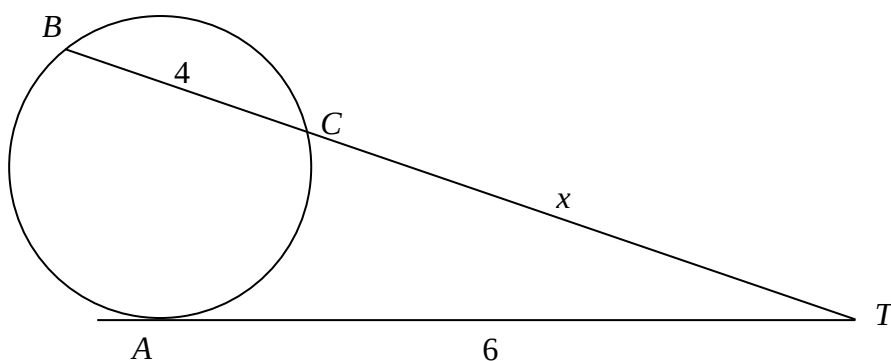
Question 2 (12 marks) Use a SEPARATE writing booklet. **Marks**

- (a) The polynomial equation $2x^3 - 3x^2 + 4x - 7 = 0$ has roots α, β and γ . **3**

Find the exact value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$.

- (b) Find $\frac{d}{dx} \cos^{-1}(4x^3)$. **2**

- (c) **2**



The line AT is the tangent to the circle at A , and BT is a secant meeting the circle at B and C .

Given that $AT = 6$, $BC = 4$ and $CT = x$, find the exact value of x .

- (d) Use the t -method to solve $5\sin x - 2\cos x = 2$, where $0^\circ \leq x \leq 360^\circ$ **3**

- (e) A debating team of three people is to be chosen from five English teachers and four Mathematics teachers.

- (i) In how many ways can the team be chosen? **1**

- (ii) What is the probability that the whole of the team will be Mathematics teachers? **1**

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int \cos^2 6x \, dx$ 2

(b) Let $P(x) = 3x^3 - ax^2 - bx + 1$, where $P(x)$ is a polynomial and a and b are real numbers. 3
When $P(x)$ is divided by $(x - 1)$ there is no remainder.
When $P(x)$ is divided by $(x + 2)$ the remainder is 15.

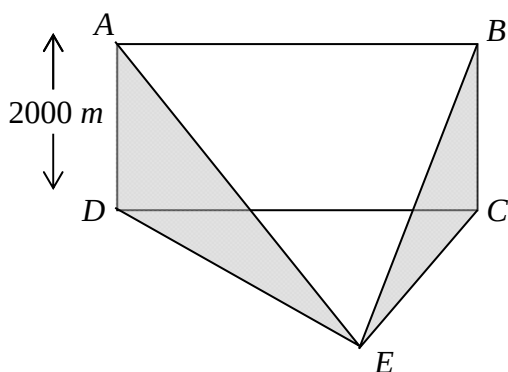
Find the values of a and b .

(c) Two parallel lines have equations $x - my + 1 = 0$ and $x - my - 1 = 0$

(i) Show that $\left(0, \frac{1}{m}\right)$ lies on $x - my + 1 = 0$. 1

(ii) Find m given that the perpendicular distance between the lines is 1 unit. 2

(d)



A plane is flying horizontally from A to B at a height of 2000m. 4
From a point E the angle of elevation of A is 15° .
From E the angle of elevation of B is 25° .

If $\angle DEC = 60^\circ$, calculate the distance AB to the nearest metre.

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Use mathematical induction to prove that **3**

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$$
- (b) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$.
- (i) Prove that the tangent at P cuts the y -axis at $T(0, -ap^2)$. **2**
- (ii) M is the midpoint of the interval PT . **2**
 Find the locus of M .
- (c) Maddison is one of ten members of the chess club. Each week one member is selected at random to win a prize.
- (i) What is the probability that in the first 7 weeks Maddison will win at least 1 prize? **1**
- (ii) Show that in the first 20 weeks, Maddison has a greater chance of winning exactly two prizes, than of winning exactly one prize. **2**
- (iii) For how many weeks must Maddison participate in the prize drawing so that she has a greater chance of winning exactly 3 prizes than of winning exactly 2 prizes. **2**

Question 5 (12 marks) Use a SEPARATE writing booklet.

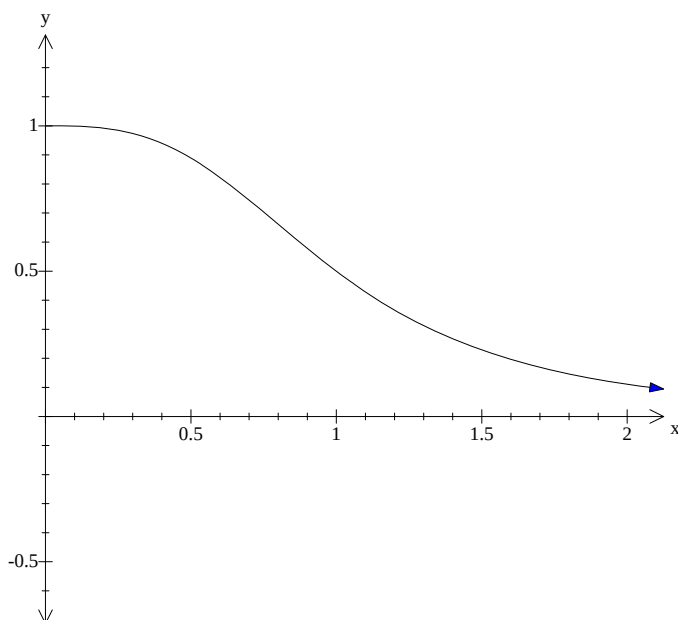
Marks

(a) (i) Evaluate $\int_2^8 \frac{8}{x} dx$, write your answer in the form $a \log 2$, where a is an integer **2**

(ii) Use Simpson's Rule with 3 function values to find an approximation for $\int_2^8 \frac{8}{x} dx$ **2**

(iii) Hence find an approximation for $\log 2$, correct to 3 decimal places. **1**

(b) The diagram below shows a sketch of the graph $y = f(x)$, where $f(x) = \frac{1}{1+x^3}$, for $x \geq 0$.



(i) Copy or trace this diagram into your answers. **1**
On the same set of axes, sketch the inverse function, $y = f^{-1}(x)$.

(ii) State the domain of $f^{-1}(x)$. **1**

(iii) Find an expression for $f^{-1}(x)$ in terms of x . **2**

(iv) The graphs of $y = f(x)$ and $y = f^{-1}(x)$ meet at exactly one point P . **1**
Let α be the x -coordinate of P . Explain why α is a root of the equation

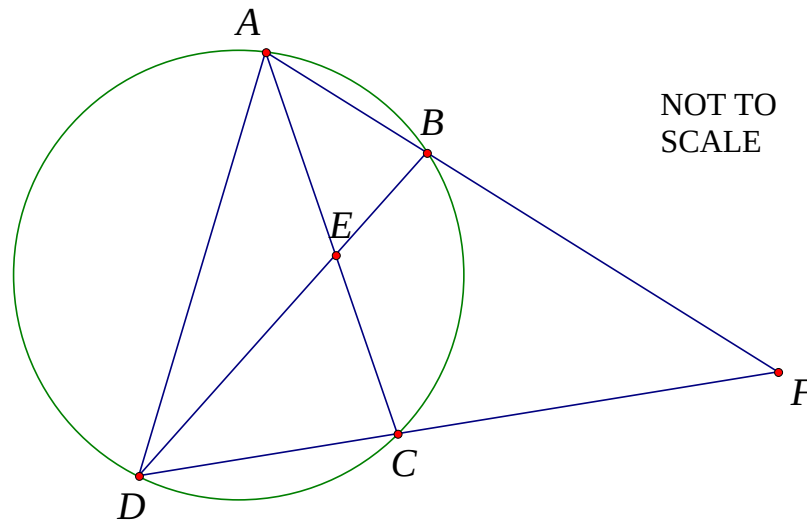
$$x^4 + x - 1 = 0.$$

(v) Take 0.5 as a first approximation to α . Use one application of Newton's method to find a second approximation for α . **2**

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



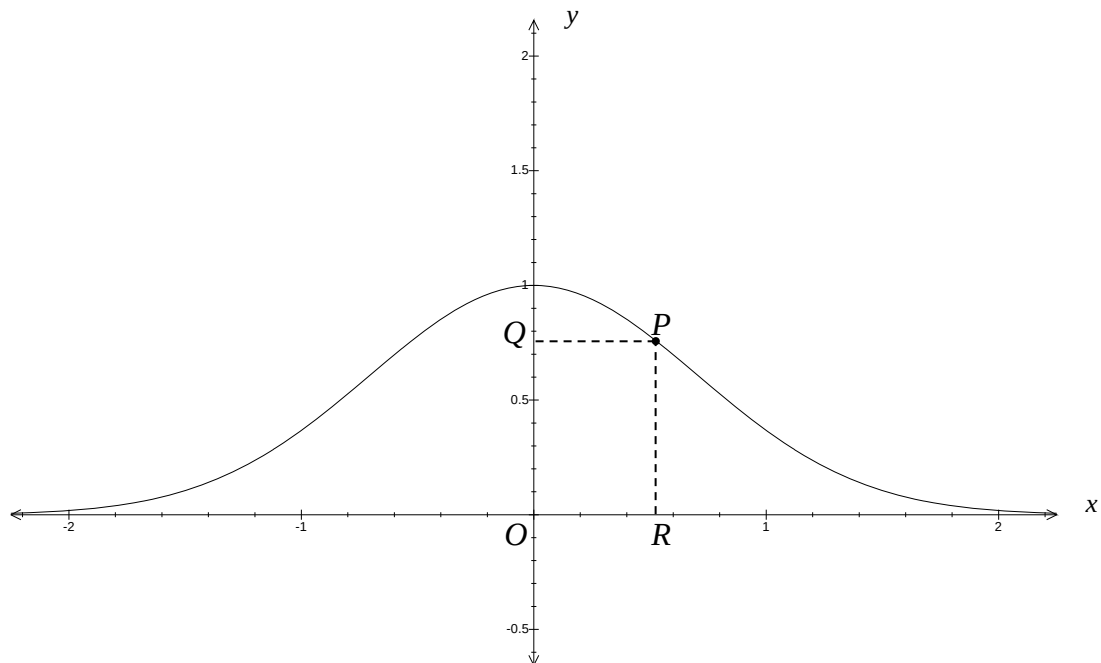
The points A , B , C and D are placed on a circle of radius r such that AC and BD meet at E . The lines AB and DC are produced to meet at F , and $BECF$ is a cyclic quadrilateral.

Copy or trace this diagram into your answer booklet.

- | | | |
|------|--|---|
| (i) | Find the size of $\angle DBF$, giving reasons for your answer. | 2 |
| (ii) | Find an expression for the length of AD in terms of r . | 2 |
| | | |
| (b) | In the expansion of $(1+ax)^9$ the coefficient of x^5 is twice the coefficient of x^6 .
Find the value of the constant a . | 4 |
| | | |
| (c) | A die is biased so that in any single throw the probability of an odd score is p ,
where p is a constant such that $0 < p < 1$, $p \neq \frac{1}{2}$. | |
| (i) | Show that in six throws of the die the probability of at most one even score
is given by $6p^5 - 5p^6$. | 2 |
| (ii) | Find the probability that in six throws of the die the product of the scores is even. | 2 |

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks



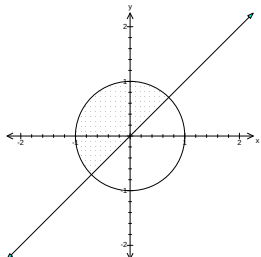
- (a) P is a point on the curve $y = e^{-x^2}$. The origin is at O and perpendiculars through P to the coordinate axes meet those axes at Q and R . Show that the maximum area of the rectangle $OQPR$ is given by $\frac{1}{\sqrt{2e}}$. 5
- (b) (i) Show that $\sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$. 2
- (ii) Prove that the derivative of $y = e^x \sin x$ is given by $\frac{dy}{dx} = \sqrt{2} e^x \sin\left(x + \frac{\pi}{4}\right)$ 2
- (iii) If $y = e^x \sin x$ prove by mathematical induction that the n^{th} derivative is given by $\frac{d^n y}{dx^n} = (\sqrt{2})^n e^x \sin\left(x + \frac{n\pi}{4}\right)$. 3

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Question	Criteria	Marks	Bands
1(a)	$\frac{4}{x+1} \geq 1, x \neq -1,$ Let $\frac{4}{x+1} = 1$ $x = 3$ $-1 \leq x \leq 3$	1 1 1	

Question	Criteria	Marks	Bands
1(b)	$\int_0^3 \frac{dx}{\sqrt{9-x^2}} = \left[\sin^{-1} \frac{x}{3} \right]_0^3$ $= \frac{\pi}{2}$	1 1	

Question	Criteria	Marks	Bands
1(c)	(-2, 1) (5, 2). 5:-4 $x = \frac{kx_2 + lx_1}{k+l}, y = \frac{ky_2 + ly_1}{k+l}$ $x = \frac{5 \times 5 + -4 \times -2}{5 + -4}, y = \frac{5 \times 2 + -4 \times 1}{5 + -4}$ $x = 33, y = 6 \rightarrow P(33, 6)$	 1 1	

Question	Criteria	Marks	Bands
1(d)	 1 for correct graphs 1 for correct shading	1 1	

Question	Criteria	Marks	Bands
1(e)	$\int_2^3 x\sqrt{x-2} dx \quad x-2 = u^2 \rightarrow x = u^2 + 2 \rightarrow dx = 2udu$ $u_1^2 = 3-2 \rightarrow u_1 = 1, u_1^2 = 2-2 \rightarrow u_2 = 0$ $\int_2^3 x\sqrt{x-2} dx = \int_0^1 (u^2 + 2)\sqrt{u^2} 2udu$ $2 \int_0^1 (u^4 + 2u^2) du = 2 \left[\frac{u^5}{5} + \frac{2u^3}{3} \right]_0^1 = \frac{26}{15}$	1 1 1	

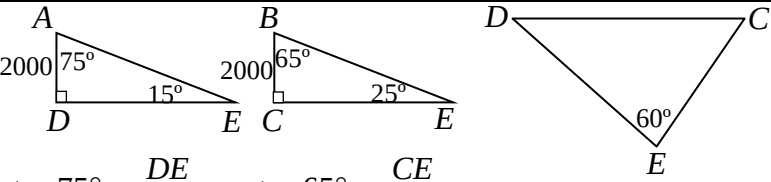
Question	Criteria	Marks	Bands
2(a)	$2x^3 - 3x^2 + 4x - 7 = 0$ $\alpha + \beta + \gamma = \frac{3}{2}, \alpha\beta + \alpha\gamma + \beta\gamma = 2, \alpha\beta\gamma = \frac{7}{2}$ $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = \frac{\gamma + \beta + \alpha}{\alpha\beta\gamma}$ $= \frac{3}{2} \div \frac{7}{2} = \frac{3}{7}$	 1 1 1	

Question	Criteria	Marks	Bands
2(b)	$\frac{d}{dx} \cos^{-1}(4x^3) = \frac{-1}{\sqrt{1-(4x^3)^2}} \times 12x^2$ $= \frac{-12}{\sqrt{1-16x^6}}$	 1 1	 one mark if only the denominator is correct

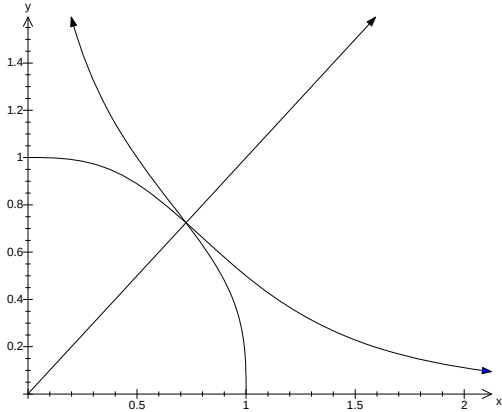
Question	Criteria	Marks	Bands
2(c)	$x(x+4) = 6^2$ (product of secants = square of the tangent) $x^2 + 4x = 36$ $(x+2)^2 = 40$ $x = -2 + 2\sqrt{10}$	 1 1	

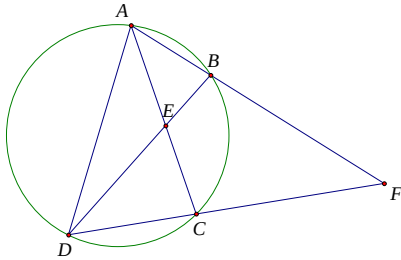
Question	Criteria	Marks	Bands
2(d)	$5 \sin x - 2 \cos x = 2 \rightarrow 5 \times \frac{2t}{1+t^2} - 2 \times \frac{1-t^2}{1+t^2} = 2$ $10t - 2 + 2t^2 = 2 + 2t^2$ $10t = 4 \rightarrow t = 0.4 \rightarrow \tan \frac{x}{2} = 0.4$ $\frac{x}{2} = 21^\circ 48', x = 43^\circ 36'$ check 180° ; $5 \sin 180^\circ - 2 \cos 180^\circ = 2 \checkmark$ $x = 43^\circ 36', 180^\circ$	 1 1 1	

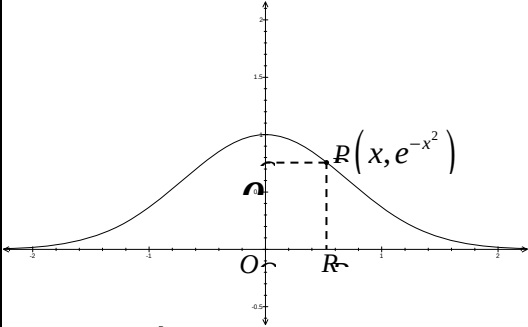
Question	Criteria	Marks	Bands
2(e)(i)	${}^9C_3 = 84$	1	
2(e)(ii)	$\frac{{}^4C_3}{{}^9C_3} = \frac{4}{84} = \frac{1}{21}$	1	

Question	Criteria	Marks	Bands
3(a)	$\cos^2 x = \frac{1}{2}(\cos 2x + 1) \rightarrow \cos^2 6x = \frac{1}{2}(\cos 12x + 1)$ $\int \cos^2 6x dx = \int \frac{1}{2}(\cos 12x + 1) dx = \frac{\sin 12x}{24} + \frac{x}{2} + C$	1 1	
3(b)	$P(x) = 3x^3 - ax^2 - bx + 1$ $P(1) = 0 \rightarrow 3 - a - b + 1 = 0 \rightarrow a + b = 4$ $P(-2) = 15 \rightarrow -24 - 4a - 2b + 1 = 15 \rightarrow -2a + b = 19$ $-2a + b = 19$ $a + b = 4$ $a = -5, b = 9$	1 1 1	
3(c)(i)	$x - my + 1 = 0$ $0 - m\left(\frac{1}{m}\right) + 1 = 0, \left(0, \frac{1}{m}\right)$ $0 = 0 \checkmark$	1	
3(c)(ii)	$x - my - 1 = 0, \left(0, \frac{1}{m}\right)$ $p.d. = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}} x - my - 1 = 0, \left(0, \frac{1}{m}\right)$ $p.d. = \frac{ 0 - 1 - 1 }{\sqrt{1^2 + (-m)^2}}$ $= \frac{2}{\sqrt{1 + m^2}} = 1 \rightarrow m = \pm\sqrt{3}$	1 1	
3(d)	 $\tan 75^\circ = \frac{DE}{2000} \quad \tan 65^\circ = \frac{CE}{2000} \quad \text{now } DC = AB$ $DE = 2000 \tan 75^\circ \quad CE = 2000 \tan 65^\circ$ $DC^2 = (2000 \tan 65^\circ)^2 + (2000 \tan 75^\circ)^2$ $-2 \times 2000 \tan 65^\circ \times 2000 \tan 75^\circ \cos 60^\circ$ $DC = 6488\text{m (nearest metre)}$ (1 mark for diagrams, 1 for relationships for DE and CE)	1 1 1 1	

[illegible]

Question	Criteria	Marks	Bands
5(a)(i)	$\int_2^8 \frac{8}{x} dx = 8[\ln x]_2^8$ $= 8(\ln 8 - \ln 2)$ $= 8 \ln 4 = 8 \ln 2^2$ $= 16 \ln 2$	1 1	
5(a)(ii)	$\int_2^8 \frac{8}{x} dx \approx \frac{8-2}{6} (f(2) + 4f(5) + f(8))$ $= 4 + 4 \times 1.6 + 1$ $= 11.4$	1 1	
5(a)(iii)	$16 \ln 2 \approx 11.4 \rightarrow \ln 2 \approx \frac{11.4}{16} = 0.7125$	1	
5(b)(i)		1	
5(b)(ii)	$x \leq 1$	1	
5(b)(iii)	$y = \frac{1}{1+x^3}$ To find the inverse function swap x and y . $x = \frac{1}{1+y^3}$ $1+y^3 = \frac{1}{x}$ $y^3 = \frac{1}{x} - 1$ $y = \sqrt[3]{\frac{1-x}{x}}$	1 1	
5(b)(iv)	A function and its inverse intersect on the line $y = x$ $\frac{1}{1+x^3} = x \rightarrow 1 = x + x^4 \rightarrow x^4 + x - 1 = 0$	1	
5(b)(v)	$f(x) = x^4 + x - 1, f'(x) = 4x^3 + 1 \rightarrow f\left(\frac{1}{2}\right) = \frac{-7}{16} \rightarrow f'\left(\frac{1}{2}\right) = \frac{3}{2}$ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{1}{2} - \frac{f\left(\frac{1}{2}\right)}{f'\left(\frac{1}{2}\right)} = \frac{1}{2} - \frac{\frac{-7}{16}}{\frac{3}{2}} = \frac{19}{24}$	1 1	

Question	Criteria	Marks	Bands
6(a)	 Let $\angle DBF = x \rightarrow \angle ECF = 180^\circ - x$ ($EBFC$ is cyclic). $\angle ECD = x$ (straight angle) $\angle ABD = x$ (subtended by arc AD) <i>but</i> $\angle EBF + \angle ABD = 180^\circ \rightarrow 2x = 180^\circ \rightarrow x = 90^\circ$	 1 1	
6(a)(ii)	From (i) $\angle DBF = 90^\circ \rightarrow \angle ABD = 90^\circ$ (straight angle) This means that $\angle ABD$ is the angle in a semi-circle. So AD is a diameter $\rightarrow$ length of $AD = 2r$.	 1 1	
6(b)	The coefficient of $x^5 = {}^9C_5 a^5 = 126a^5$ The coefficient of $x^6 = {}^9C_6 a^6 = 84a^6$ $126a^5 = 2 \times 84a^6$ $a = \frac{3}{4}$	 1 1 1 1	
6(c)(i)	$P(\text{at most one even}) = P(\text{none even}) + P(\text{one even})$ $= p^6 + {}^6C_1 p^5 (1-p)$ $= p^6 + 6p^5 - 6p^6$ $= 6p^5 - 5p^6$	 1 1	
6(c)(ii)	For the product to be odd – all scores would be odd. $P(\text{all odd}) = p^6 \rightarrow P(\text{even product}) = 1 - p^6$	 1 1	

Question	Criteria	Marks	Bands								
7	 $A = x \times e^{-x^2}$ $A' = e^{-x^2} \cdot 1 + x \cdot -2xe^{-x^2}$ $= e^{-x^2} (1 - 2x^2)$ Maximum area occurs when $A' = 0$; $e^{-x^2} (1 - 2x^2) = 0$ $e^{-x^2} = 0, \quad 1 - 2x^2 = 0$ no solution $x = \frac{1}{\sqrt{2}}$ (since P is in the 1st quadrant) <table border="1" data-bbox="316 974 571 1055"> <tr> <td>x</td><td>0</td><td>$1/\sqrt{2}$</td><td>1</td></tr> <tr> <td>A'</td><td>+</td><td>0</td><td>-</td></tr> </table> So this is a maximum area. $A = \frac{1}{\sqrt{2}} e^{-\left(\frac{1}{\sqrt{2}}\right)^2}$ $= \frac{1}{\sqrt{2}} e^{-\frac{1}{2}} = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{e}} = \frac{1}{\sqrt{2e}}$	x	0	$1/\sqrt{2}$	1	A'	+	0	-	1 1 1 1 1	
x	0	$1/\sqrt{2}$	1								
A'	+	0	-								
7(b)(i)	$\sin x + \cos x = A \sin(x + \alpha)$ $\sin x + \cos x = A \sin x \cos \alpha + A \cos x \sin \alpha$ $A \cos \alpha = 1, A \sin \alpha = 1 \rightarrow \frac{A \sin \alpha}{A \cos \alpha} = 1 \rightarrow \tan \alpha = 1 \rightarrow \alpha = \frac{\pi}{4}$ $A = \sqrt{1^2 + 1^2} = \sqrt{2} \rightarrow \sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right).$	1 1									
7(b)(ii)	$y = e^x \sin x \rightarrow y' = \sin x \cdot e^x + e^x \cdot \cos x = e^x (\cos x + \sin x)$ $\text{but } \sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$ $y' = e^x \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) = \sqrt{2} e^x \sin\left(x + \frac{\pi}{4}\right)$	1 1									

7(b)(iii)	RTP $\frac{d^n y}{dx^n} = (\sqrt{2})^n e^x \sin\left(x + \frac{n\pi}{4}\right)$. when $n = 1$: $\frac{d^n y}{dx^n} = (\sqrt{2})^n e^x \sin\left(x + \frac{n\pi}{4}\right) \rightarrow \frac{dy}{dx} = (\sqrt{2})^1 e^x \sin\left(x + \frac{1\pi}{4}\right)$ true from (ii) suppose it is true for some value of n say k. $\frac{d^k y}{dx^k} = (\sqrt{2})^k e^x \sin\left(x + \frac{k\pi}{4}\right).$ Prove it is true for $n = k + 1$ RTP $\frac{d^{k+1} y}{dx^{k+1}} = (\sqrt{2})^{k+1} e^x \sin\left(x + \frac{(k+1)\pi}{4}\right)$. $\frac{d^{k+1} y}{dx^{k+1}} = \frac{d}{dx} \frac{d^k y}{dx^k} = \frac{d}{dx} (\sqrt{2})^k e^x \sin\left(x + \frac{k\pi}{4}\right).$ $= (\sqrt{2})^k \left(\sin\left(x + \frac{k\pi}{4}\right) e^x + e^x \cos\left(x + \frac{k\pi}{4}\right) \right)$ $= (\sqrt{2})^k e^x \left(\sin\left(x + \frac{k\pi}{4}\right) + \cos\left(x + \frac{k\pi}{4}\right) \right)$ $= (\sqrt{2})^k e^x \sqrt{2} \sin\left(x + \frac{k\pi}{4} + \frac{\pi}{4}\right) \text{ from (i)}$ $= (\sqrt{2})^{k+1} e^x \sin\left(x + \frac{(k+1)\pi}{4}\right)$ proved by induction.	1	
		1	
		1	